Socket_Burners

PulseView

Overview

- 3 Lead ECG
- Input Signal 2–4 mV range
- Variable Gain and Q-Factor
- Portable with 10+ hours battery life

1 Description

- PulseView is a portable battery-powered ECG device
- Three-electrode configuration (RA, LA, RL)
- Signal acquisition stages:
  - Instrumentation Amplifier
  - High-Pass Filter (0.5 Hz)
  - Low-Pass Filter (100 Hz)
  - Notch Filter (50 Hz)
- Right Leg Drive for common-mode rejection
- Output suitable for oscilloscope viewing

2 Technical Specification

General Specifications	
Power Voltage	3.7 V
Active Current Consumption	35 mA
CMRR	120 dB
Input Impedance	$10\,\mathrm{M}\Omega$
Output Impedance	$1\,\mathrm{k}\Omega$
ECG Signal Specifications	
Number of Leads	3
Frequency Range	0.5 – 100 Hz
Maximum Safe Input Voltage	6mV
Physical Specifications	
Dimensions	146mm x 99mm x 71mm
Weight	50 g (no batteries) / 250 g (with batteries)
Enclosure Material	PLA Plastic
Connector Type	3.5 mm Jack
Power Specifications	
Battery Type	3.7 V Li-Ion, 3200 mAh $\times 4$
Battery Life	10+ hours
Nominal Voltage	± 7.4 V
Operating Voltage	± 8.2 V

3 Printed Circuit Board

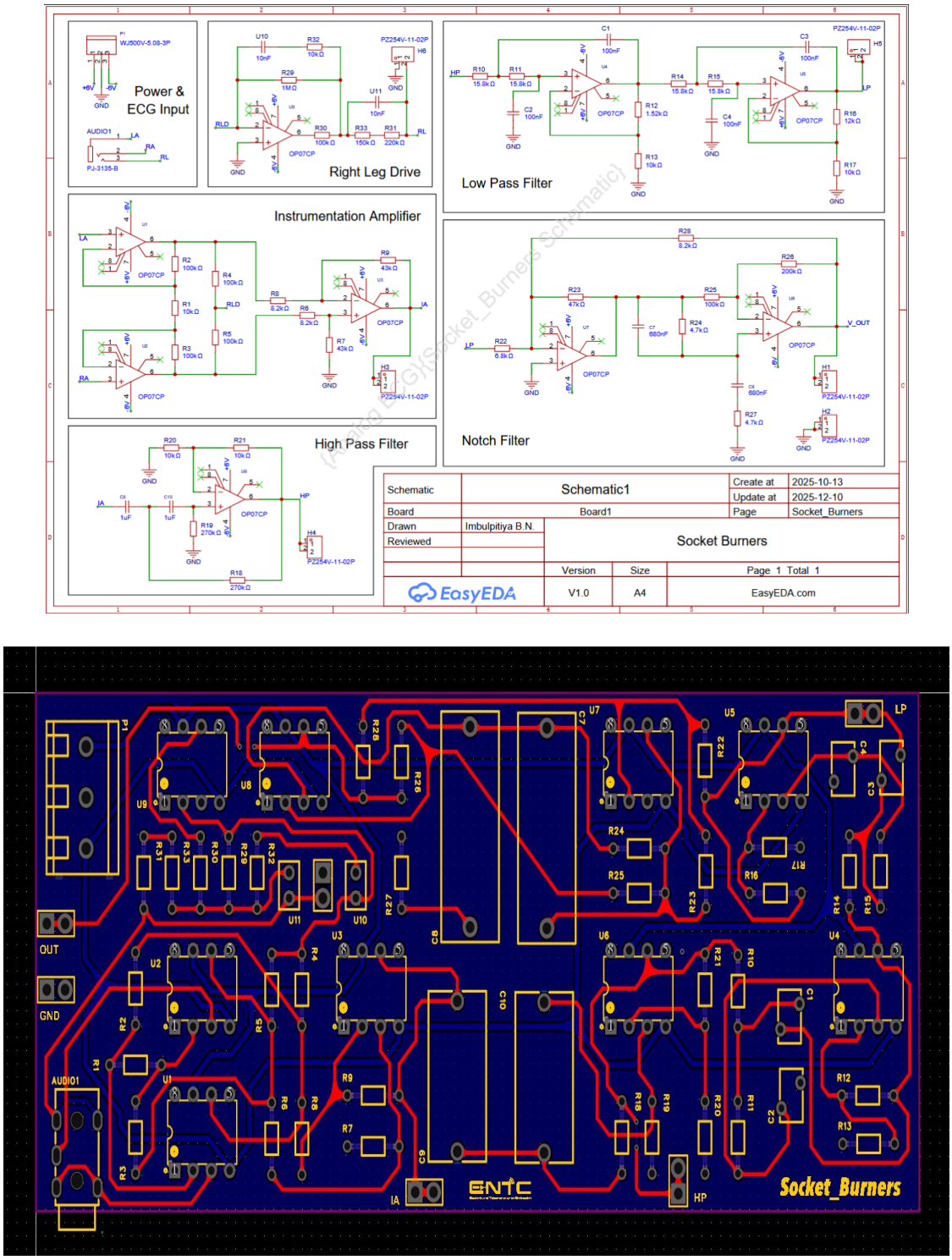


Figure 1: PCB schematic and fabricated PCB with dimensions

Quantity	Comment	Designator
1	PJ-3135-B	AUDIO1
4	100nF	C1,C2,C3,C4
2	680nF	C7,C8
2	1uF	C9,C10
6	PZ254V-11-02P	H1,H2,H3,H4,H5,H6
1	WJ500V-5.08-3P	P1
6	10kΩ	R1,R13,R17,R20,R21,R32
6	100kΩ	R2,R3,R4,R5,R25,R30
3	8.2kΩ	R6,R8,R28
2	43kΩ	R7,R9
4	15.8kΩ	R10,R11,R14,R15
1	1.52kΩ	R12
1	12kΩ	R16
2	270kΩ	R18,R19
1	6.8kΩ	R22
1	47kΩ	R23
2	4.7kΩ	R24,R27
1	200kΩ	R26
1	1MΩ	R29
1	220kΩ	R31
1	150kΩ	R33
9	OP07CP	U1,U2,U3,U4,U5,U6,U7,U
2	10nF	U10,U11

Figure 2: List of Components

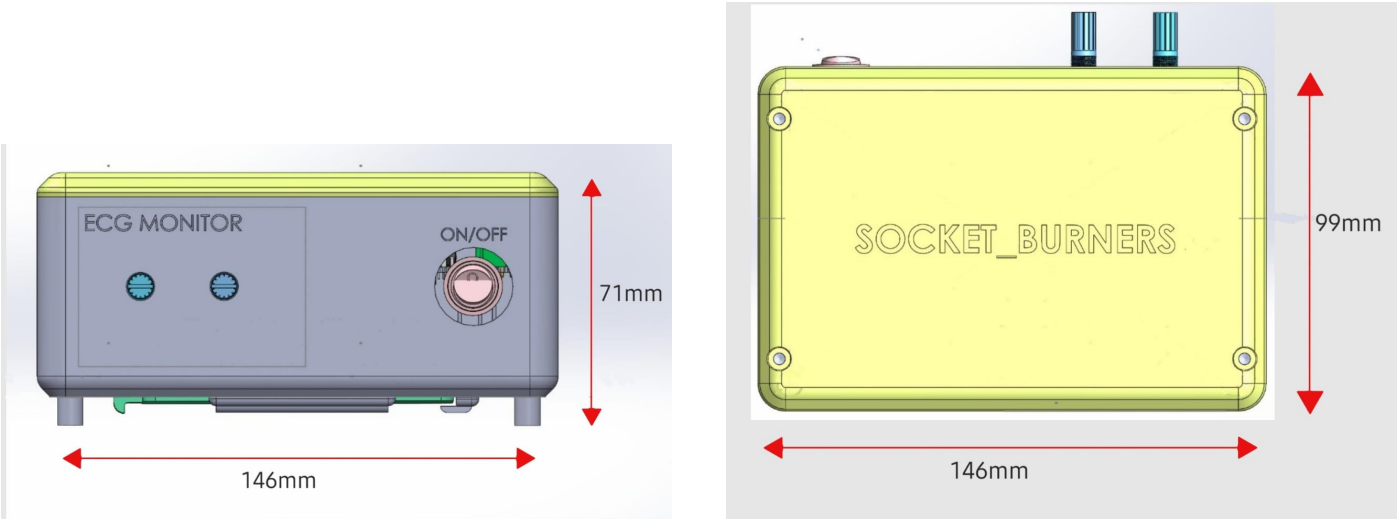
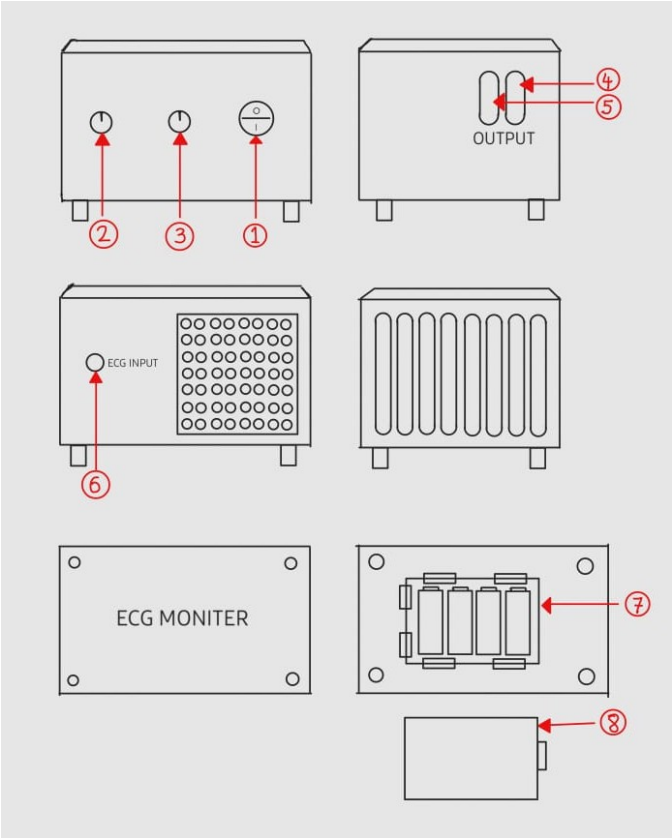


Figure 3: Device Dimensions



No.	Component Description
1	On/Off switch
2	Instrumentation amplifier gain adjust
3	Notch filter adjustment
4	Output signal port
5	Output ground port
6	Auxiliary ECG input
7	Battery compartment
8	Battery cover
9	ECG auxiliary port
10	Red lead (LA)
11	Green lead (RA)
12	Yellow lead (LL)

Figure 4: Labelled overview of the ECG device

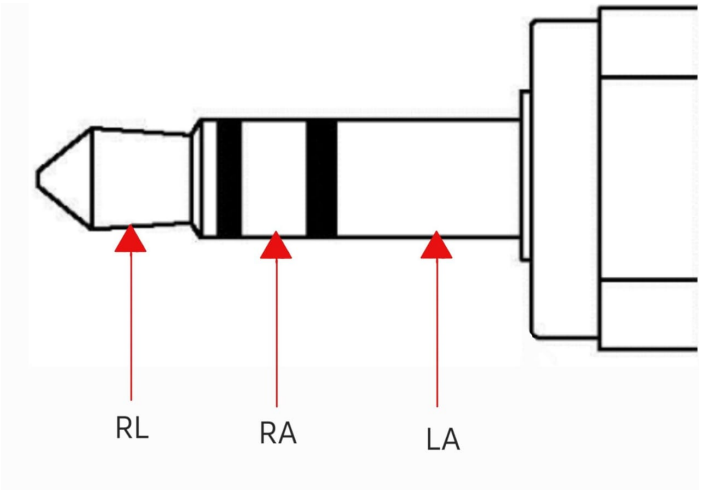


Figure 5: Labelled Diagram of Input Connector